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Electronic apparatus and control method of the same

Abstract

An electronic apparatus a determining unit configured to determine a target object that a user is paying attention to, a detecting unit configured to detect a plurality of types of detection information on the target object, a referring unit configured to refer to reference information associated with the target object, a presenting unit configured to present at least one of the plurality of types of detection information, and a control unit configured to determine the at least one of the plurality of types of detection information presented by the presenting unit, based on the reference information and the plurality of types of detection information.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to an electronic apparatus and a control method of the same.

Description of the Related Art

(2) A work support system using a wearable terminal such as a head mount display (“HMD”) presents information on a work target to a display screen, and thus a user can acquire necessary information without taking his/her hand off the work target. For example, Japanese Patent Laid-Open No. (“JP”) 2018-74200 discloses a drawing projection system and method for superimposing and displaying three-dimensional drawing data of a construction site on a see-through type display screen of the wearable terminal. For example, in the inspection step, the user can work while comparing the drawing data and the actual work target with each other, and find a defect of the work target etc. at an early stage, improving the accuracy and efficiency of the inspection work.

(3) As an applied work support system, for example, a wearable terminal includes a plurality of different sensing devices, such as a position detecting device and a nonvisible light wavelength detecting device in addition to a camera, and presents various work support information based on the sensing results. This wearable terminal can present to the display screen visually unnoticeable, useful sensing information on the work target, and is expected to further improve the work efficiency.

(4) However, it is annoying to always display various information on the work target on the display screen. Accordingly, for example, JP 2015-207168 discloses a method of recognizing an object from a captured image and of presenting information corresponding to the recognized object.

(5) The method disclosed in JP 2015-207168 fixes information (detection information) to be displayed for each recognized target (target object). The information required by the user differs according to the state of the work target (target object). For example, it is demanded to display dimension information in a case where the dimension of the work target is a feature, and to display temperature information in a case where the temperature of the work target is a feature. However, JP 2015-207168 is silent about a method of selecting and displaying necessary information from among a plurality of types of detection information.

SUMMARY OF THE INVENTION

- (6) The present invention provides an electronic apparatus and a control method of the same, each of which can present proper detection information according to a state of a target object.
- (7) An electronic apparatus according to one aspect of the present invention includes at least one processor, and at least one memory coupled to the at least one processor storing instructions that, when executed by the at least one processor, cause the at least one processor to function as a determining unit configured to determine a target object that a user is paying attention to, a detecting unit configured to detect a plurality of types of detection information on the target object, a referring unit configured to refer to reference information associated with the target object, a presenting unit configured to present at least one of the plurality of types of detection information, and a control unit configured to determine the at least one of the plurality of types of detection information presented by the presenting unit, based on the reference information and the plurality of types of detection information. A control method corresponding to the above electronic apparatus also constitutes another aspect of the present invention.
- (8) Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is an external view of a wearable terminal according to each embodiment.
- (2) FIG. 2 is a block diagram of the wearable terminal according to each embodiment.
- (3) FIG. 3 is a block diagram of an audio signal processing unit according to each embodiment.
- (4) FIGS. 4A and 4B are configuration diagrams of an image sensor according to each embodiment.
- (5) FIGS. 5A and 5B are explanatory diagrams of a distance measurement method according to each embodiment.
- (6) FIGS. 6A to 6D explain a system that provides sensing information by a wearable terminal according to a first embodiment.
- (7) FIGS. 7A and 7B illustrate a flowchart of a method of providing sensing information by the wearable terminal according to the first embodiment.
- (8) FIG. 8 is a table for storing sensing information by the wearable terminal according to the first embodiment.
- (9) FIGS. 9A to 9C explain a system that supports a user having a blind spot using two wearable terminals according to a second embodiment.
- (10) FIGS. 10A and 10B illustrate flowcharts of a method of supporting the user having the blind spot using two wearable terminals according to the second embodiment.
- (11) FIG. 11 is a display example of positional relationship information shared between the two wearable terminals according to the second embodiment.
- (12) FIG. 12 is a flowchart of a method of measuring dimension using a wearable terminal according to a third embodiment.
- (13) FIGS. 13A and 13B explain the distance measurement reliability of two distance measurement units mounted on the wearable terminal according to the third embodiment.
- (14) FIG. 14 explains a relationship between an imaging angle of view and a display angle of view of a wearable terminal according to a fourth embodiment.
- (15) FIG. 15 is a bird's-eye view of the relationship between the imaging angle of view and the display angle of view of the wearable terminal according to the fourth embodiment.
- (16) FIGS. 16A to 16F explain display angle of view control for a translational movement of the wearable terminal according to the fourth embodiment.
- (17) FIGS. 17A to 17D explain the display angle of view control for an optical-axis rotation of the wearable terminal according to the fourth embodiment.

(18) FIG. 18 is a flowchart of the display angle of view control by the wearable terminal according to the fourth embodiment.

DESCRIPTION OF THE EMBODIMENTS

(19) Referring now to the accompanying drawings, a detailed description will be given of embodiments according to the present invention.

(20) Referring now to FIG. 1, a description will be given of a wearable terminal (electronic apparatus) **100** according to each embodiment. FIG. 1 is an external view of the wearable terminal **100**. In FIG. 1, a right-eye imaging unit **150R** and a left-eye imaging unit **150L** are imaging unit sections including a lens (lens unit) **103** and an imaging unit **22**, which will be described below, respectively. Right-eye and left-eye captured images can be displayed on a right-eye display unit **160R** and a left-eye display unit **160L**, which will be described below, respectively. Each of the imaging unit **150R** and the imaging unit **150L** includes a zoom mechanism, which can change a zoom magnification using an operating unit such as a voice UI (user interface) described below. The zoom operation can combine an optical zoom by a lens zoom mechanism and an electronic zoom by cutting out a captured image.

(21) An infrared LED illumination unit **81** and a TOF (Time of Flat) detecting unit **83** (collectively referred to as a second distance measurement unit) are a light emitting illumination unit and a detecting unit, respectively, of a distance measurement system using the TOF type. However, the type of this embodiment is not limited to this example, and other methods such as a triangulation type may be used. The infrared LED illumination unit **81** is disposed in a ring shape about the TOF detecting unit **83**, and can irradiate an area much wider than an angle of view of the TOF detecting unit **83**. The TOF type distance measurement principle will be described below.

(22) Each of the right-eye display unit **160R** and the left-eye display unit **160L** is a display unit section including an eyepiece unit **16**, a display unit (presenting unit) **29**, and a visual-line detecting unit **41**, which will be described below. The eyepiece unit **16** is an eyepiece unit for an eyepiece finder (a peep type finder), and the user can visually recognize an image (video) displayed on the internal display unit **29** via the eyepiece unit **16**.

(23) Referring now to FIG. 2, a description will be given of an internal configuration of the wearable terminal **100**. FIG. 2 is a block diagram of the wearable terminal **100**. Although the lens **103** includes a plurality of lenses, it is simply illustrated as a single lens in FIG. 2. A system control unit **50** realizes diaphragm control by communicating with a lens system control circuit **4** and controlling a diaphragm (aperture stop) **1** via a diaphragm driving circuit **2**. A focus lens in the lens **103** is displaced via an AF driving circuit **3** to focus on the object.

(24) A dichroic mirror **21** is disposed for a wavelength separation, and has a structure that transmits the visible light while reflecting the infrared light. The visible light component that has transmitted through the dichroic mirror **21** is photoelectrically converted by the imaging unit **22** for the visible light disposed subsequent to the dichroic mirror **21**, and a visible light image signal is generated. The infrared light component reflected by the dichroic mirror **21** is photoelectrically converted by an infrared imaging unit **25** to generate an infrared light image signal.

(25) The imaging unit **22** has an image sensor that includes a CCD, a CMOS element, or the like, and converts an optical image into an electric signal. The imaging unit **22** includes, for example, a single-plate color image sensor for imaging visible light provided with a general primary color filter. The primary color filter includes three types of color filters having main transmission wavelength bands near 650 nm, 550 nm, and 450 nm, respectively, and captures images corresponding to the R (red), G (green), and B (blue) bands, respectively. The imaging unit **22** includes a built-in A/D converter and is used to convert an analog signal output from the image sensor into a digital signal. The imaging unit **22** has a pixel configuration capable of a distance measurement (focus detection) by an imaging-plane phase-difference distance measurement type (imaging-plane phase-difference detecting type) described below.

(26) Similar to the imaging unit **22**, the infrared imaging unit **25** has an image sensor that includes a

CCD, a CMOS element, or the like and converts an optical image into an electric signal, and may use an image sensor made of Si, InGaAs, InAsSb, or the like, depending on the wavelength band. Thereby, near-infrared light (780 to 2500 nm), mid-infrared light (2500 to 25000 nm), and far-infrared light (4 μ m to 1000 μ m) are detectable. The near-infrared light is characterized in that its wavelength is not easily absorbed by objects and has high transmitting performance. Utilizing this characteristic, images captured with the near-infrared light are suitable to determine scratches and foreign matters (dirt) on the object. The mid-infrared light or far-infrared light can be used as a thermography to measure the temperature of an object.

(27) An image processing unit **24** performs resizing processing, such as predetermined pixel interpolation and reduction, and color conversion processing for data from the imaging unit **22** or the infrared imaging unit **25** or data from a memory control unit **15** described below. The image processing unit **24** performs predetermined calculation processing using captured image data. The system control unit **50** performs exposure control and distance measurement control based on the calculation result obtained by the image processing unit **24**. Thereby, a TL (through-the-lens) AF (autofocus) processing and AE (auto-exposure) processing are performed. The image processing unit **24** performs predetermined calculation processing using the captured image data, and performs TTL type AWB (auto white balance) processing based on the obtained calculation result.

(28) The memory control unit **15** controls data transmissions and receptions among the imaging unit **22**, the image processing unit **24**, and a memory **32**. The output data from the imaging unit **22** is written down to the memory **32** via the image processing unit **24** and the memory control unit **15** or directly via the memory control unit **15**. The memory **32** stores the image data obtained by the imaging unit **22** and the image data to be displayed on the display unit **29**. The memory **32** also serves as a memory (video memory) for displaying an image. The image data to be displayed written down on the memory **32** is displayed by the display unit **29** via the memory control unit **15**. The display unit **29** displays on a display device such as an LCD or an organic EL according to a signal from the memory control unit **15**. Sequentially transferring the image data stored in the memory **32** to the display unit **29** and displaying the image data there can provide live-view display of the captured image.

(29) A visual-line detecting unit **41** detects the user's visual-line positions in the viewfinder screen using an unillustrated infrared light emitting diode. The irradiated infrared light is reflected by the user's eyeballs (eyes) **161R** and **161L**, respectively, and the infrared reflected light reaches the dichroic mirror **42**. The dichroic mirror **42** reflects infrared light and transmits visible light. The infrared reflected light whose optical path has been changed is imaged on the imaging plane of the visual-line detecting unit **41** through an unillustrated imaging lens. The visual-line detecting unit **41** includes an image sensor such as a CCD image sensor.

(30) A gaze determining unit **43** determines whether or not the user is gazing at (paying attention to) a specific imaging target (object). In a case where the detection result is received from the visual-line detecting unit **41** and the time when the user's visual-line is fixed to a point on an arbitrary object exceeds a predetermined threshold, it is determined that the user is gazing at that point. An object corresponding to a gaze point determined by the gaze determining unit **43** will be referred to as a gazed object (target object).

(31) A nonvolatile memory **56** is an electrically erasable and recordable memory, and uses, for example, a Flash-ROM or the like. The nonvolatile memory **56** stores a constant, a program, and the like for the operation of the system control unit **50**. The program, as used herein, means a program for executing various flowcharts described below according to each embodiment.

(32) The system control unit **50** is a control unit including at least one processor or circuit, and controls the entire wearable terminal **100**. Executing the program recorded in the nonvolatile memory **56** enables each processing of each embodiment to be implemented as described below. For example, a system memory **52** uses a RAM that loads a constant and a variable for the operation of the system control unit **50**, a program that is read out of the nonvolatile memory **56**,

and the like. The system control unit **50** also provides display control by controlling the memory **32**, the display unit **29**, and the like. A system timer **53** is a time measuring (or timekeeping) unit configured to measure time for various controls and time of a built-in clock.

(33) An operating unit **70** includes various operating members as an input unit that accepts an operation from the user. The operating unit **70** includes, for example, a voice UI operating unit and a touchpad. FIG. **3** is a block diagram of a voice UI operating unit (voice signal processing unit **204**). A voice input unit **104** includes a microphone, and receives the voice of the user wearing the wearable terminal **100** as a soundwave and converts it into a voice signal. The voice signal processing unit **204** includes a sound pressure level detecting unit **241**, a voice memory **242**, and a voice recognizing unit **243**. The sound pressure level detecting unit **241** detects that an output level of the voice input unit **104** is equal to or higher than a certain threshold. The voice memory **242** stores voice data for commands in advance. The voice memory **242** is also used to sequentially store the output data of the voice input unit **104**, triggered by the sound pressure level detecting unit **241**. The voice recognizing unit **243** matches the specific voice command stored in the voice memory **242** with the sequentially stored voice data, and determines whether or not the voice command and the voice data coincide with each other. For example, the zoom magnification of the imaging units **150R** and **150L** can be changed by using the voice command.

(34) The touchpad is mounted on an unillustrated side surface of the wearable terminal **100**. The system control unit **50** can detect an operation on or a state of the touchpad. A position coordinate that a finger touches on the touchpad is sent to the system control unit **50** through an internal bus, and the system control unit **50** determines what kind of operation (touch operation) on the touchpad based on the sent information. The touchpad can use any of various touch panels such as a resistance film type, a capacitance type, a surface acoustic wave type, an infrared type, an electromagnetic induction type, an image recognition type, and an optical sensor type.

(35) A power switch **71** is an operating member that switches powering-on and off of the wearable terminal **100**. A power supply control unit **80** includes a battery detecting circuit, a DC-DC converter, a switching circuit for switching a block to be electrified, and the like, and detects whether or not a battery is installed, a battery type, and a remaining amount of the battery. The power supply control unit **80** controls the DC-DC converter based on the detection result and the instruction of the system control unit **50**, and supplies a necessary voltage to each component including the recording medium **180** for a necessary period. A power supply unit **30** includes a primary battery such as an alkaline battery and a lithium battery, a secondary battery such as a NiCd battery, a NiMH battery, and a lithium-ion battery, an AC adapter, and the like. A recording medium I/F18 is an interface with a recording medium **180** such as a memory card or a hard disk drive. The recording medium **180** is a recording medium such as a memory card for recording a captured image, and includes a semiconductor memory, a magnetic disk, or the like.

(36) A communication unit (referring unit) **54** is connected wirelessly or by a wire cable to transmit and receive video and audio signals. It can also transmit and receive data to and from an external database server **1010**. For example, the database server **1010** may store three-dimensional CAD data of a work target so as to construct a drawing projection system that superimposes and displays the three-dimensional drawing data relating to the work target on the display unit **29** with the actual image. The communication unit **54** is also connectable with a wireless LAN (Local Area Network) or the Internet. The communication unit **54** can also communicate with an external device using Bluetooth (registered trademark) or Bluetooth Low Energy.

(37) An orientation (or attitude) detecting unit **55** detects an orientation of the wearable terminal **100** relative to the gravity direction. The orientation detecting unit **55** can use an acceleration sensor, a gyro sensor, or the like. The orientation detecting unit **55** can detect a motion of the wearable terminal **100** (whether or not the wearable terminal **100** is panned, tilted, rolled, stands still, etc.).

(38) A wear detecting unit **57** is a wear detecting sensor that detects whether or not the wearable

terminal **100** is worn by the user. The system control unit **50** can switch starting (powering-on) and stopping (powering-off) of the wearable terminal **100** according to the state detected by the wear detecting unit **57**. The wear detecting unit **57** can use, for example, an infrared proximity sensor, and may be configured to detect an approach of some object to the eyepiece unit **16**. In a case where an object approaches, the infrared light projected from a light emitter (not shown) of the wear detecting unit **57** is reflected and received by the light receiving unit (not shown) of the infrared proximity sensor. A received infrared light amount can also be used to determine how close the object is to the eyepiece unit **16**. The infrared proximity sensor is an example, and another sensor such as a capacitance type may be adopted.

(39) An infrared LED illumination unit **81** is an irradiation unit for the TOF system, and the light is modulated to a certain frequency by the LED driving unit **82** and irradiated. The irradiated light from the infrared LED illumination unit **81** illuminates the measurement target. The reflected light from the measurement target is imaged by an imaging lens (not illustrated) and reaches the TOF detecting unit **83** with a time lag t corresponding to the measurement target distance. For example, in a case where the time lag t is 10 nanoseconds, the object distance can be calculated as 3 m (=10 nanoseconds (0.00000001) seconds $\times$ 300.000 km) because the light speed c is 300,000 km/sec. A distance measurement resolution d in the TOF detection method is expressed by the following expression (1).

$$(40) \quad d_{\text{TOF}} = c / (2 \cdot f_{\min} \cdot \sqrt{N}) \quad (1)$$

where $f_{\min}$ is a modulation frequency, and N is the number of effective signal electrons usable after noise is removed. That is, the distance measurement resolution depends on the SNR representing the signal-to-noise ratio. If measurement targets have the same reflectance, the farther the measurement target is, the smaller the SNR becomes. As a result, within a distance measurable range by the TOF detecting unit **83**, the farther the distance measurement target is, the lower the distance measurement resolution is. The distance measurable range depends on the output of the infrared LED illumination unit **81**, and a high output needs a large LED and high power consumption. Each embodiment assumes the configuration suitable for the wearable terminal, and the specification that can provide a distance measurement up to a target several meters away.

(41) A feature-point extracting unit **33** extracts a predetermined number of feature points from the image data calculated by the image processing unit **24**. The memory **32** stores information on the feature points extracted for each imaging frame by the feature-point extracting unit **33**. The feature-point extracting method can use various methods such as Scale-Invariant Feature Transfer (SIFT) and Features from Acknowledged Segment Test (FAST). Alternatively, Oriented FAST and Rotated BRIEF (ORB) may be used. The detected feature points are stored as a feature amount descriptor for calculating a binary vector capable of memory-saving and high-speed matching such as Binary Robust Independent Elementary Features (BRIEF).

(42) A feature-point tracking unit **34** reads out the feature point information stored in the memory **32** and compares it with the feature points extracted in the image data in a new imaging frame. Thereby, feature points are matched between a plurality of images. The matching method can use, for example, a brute-force matcher, a FLAN based matcher, and a Mean-Shift searching method.

(43) A three-dimensional shape restoring unit **35** restores a three-dimensional shape of an imaging scene from a plurality of acquired captured image data. The three-dimensional shape restoring unit **35** can use, for example, a method such as SfM (Structure from Motion) or SLAM (Simultaneous Localization and Mapping). In either method, the three-dimensional shape restoring unit **35** matches the extracted feature points by the feature-point tracking unit **34** with a plurality of captured images having different orientations and positions, and finds corresponding points between the images. For example, by using an optimization method, such as Bundle Adjustment, a three-dimensional position (three-dimensional points data), a camera position, and a camera orientation of the matched feature points are estimated. The three-dimensional shape restoring unit

35 may improve the estimation accuracy of the three-dimensional points data and the camera position and orientation by using a method such as Visual Initial SLAM that adds the output of the orientation detecting unit **55** to the optimization condition. The system control unit **50** may further add the distance measurement result by the TOF detecting unit **83** to improve the estimation accuracy by the SLAM.

(44) A three-dimensional shape matching unit (calculating unit) **36** aligns the shape of the three-dimensional points data generated by the three-dimensional shape restoring unit **35** and the shape of the target object on the three-dimensional CAD data stored in the external database server **1010** or the like with each other, and matches (compares) the arrangement. The alignment can use a method such as an ICP (Iterative Close Point) algorithm. Data is transmitted to and received from the external database server **1010** via the communication unit **54**. This configuration can verify which coordinate position each feature point extracted by the feature-point extracting unit **33** corresponds to in the target object coordinate system. The position and orientation of the wearable terminal **100** during imaging each image data can be estimated from the arrangement relationship.

(45) A scratch/dirt detecting unit **37** detects whether or not there is a defect due to any scratches on the measurement target or dirt caused by the adhesion of foreign matters from the infrared image data acquired by the infrared imaging unit **25**. The scratch detecting method can use, for example, a difference in infrared emissivity between the defective portion and the surface of the non-defective portion. This emissivity difference appears as a gradation difference on the infrared image data. Image processing such as edge processing and background noise cut is applied to the infrared image data, and if the length and area of the detected defect are equal to or larger than the reference determination threshold, it is determined as a defect. For example, a method of detecting scratches or foreign matter from the image data acquired by the imaging unit **22** may be used.

(46) A temperature measuring unit (temperature detecting unit) **38** measures a two-dimensional temperature distribution of the measurement target from the infrared image data acquired by the infrared imaging unit **25**. Given an ideal infrared radiator, a relationship between the radiation and the temperature follows the Stefan-Boltzmann's law. The infrared radiance emitted from the measurement target is compared with the blackbody radiance, which is an ideal infrared radiator, and can be converted into the temperature.

(47) A dimension measuring unit (dimension detecting unit) **39** measures a dimension (size) of the measurement target based on the distance measurement information acquired, for example, by the TOF detecting unit **83**. A map of an object on a captured image can be restored to a real-scale three-dimensional shape using a camera parameter that represents a known geometric relationship for perspective projecting and converting a three-dimensional coordinate point into a captured image space. This configuration can measure a dimension of a designated part of the object. The dimension measurement may utilize a restoring method using the three-dimensional shape restoring unit **35**. The distance measurement unit may use an imaging-plane phase-difference focus-detecting type of the imaging unit **22**.

(48) Referring now to FIGS. **4A** and **4B**, a description will be given of a configuration of the image sensor in the imaging unit **22**. FIGS. **4A** and **4B** are configuration diagrams of the image sensor. As illustrated in FIG. **4A**, in the image sensor, a plurality of pixels **200** are two-dimensionally and regularly arranged. More specifically, the plurality of pixels **200** is arranged, for example, in a two-dimensional lattice pattern. The arrangement of the pixels **200** is not limited to the lattice pattern, and another arrangement may be adopted.

(49) In order to realize a focus detecting function of the imaging-plane phase-difference focus-detecting type, each pixel **200** includes a microlens **201** illustrated in FIG. **4B** and pupil dividing pixels **202a** and **202b** which are a pair of photoelectric converters. The pupil dividing pixels **202a** and **202b** are photoelectric converters having the same shape and the same size, in which the imaging plane has a rectangular shape with a y-axis direction as the longitudinal direction. In each pixel **200**, the pupil dividing pixels **202a** and **202b** are arranged line-symmetrically with a

perpendicular bisector along the y-axis direction of the microlens **201** as an axis of symmetry. The shapes of the pupil dividing pixels **202a** and **202b** on the imaging plane are not limited to this example, and may be other shapes. The arrangement of the pupil dividing pixels **202a** and **202b** is not limited to the arrangement parallel to the x-axis direction, and another arrangement may be adopted.

(50) This configuration enables the imaging unit **22** according to each embodiment to output an A image relating to the image signal output from the pupil dividing pixel **202a** included in all pixels in the image sensor, and a B image relating to the image signal output from the pupil dividing pixel **202b** in the same manner. The A image and the B image have a parallax relationship according to a distance from the in-focus position.

(51) The system control unit **50** calculates a defocus amount of an image of each object included in the captured image as information on a distance distribution of the object in the depth direction of the imaging range. The defocus amount is calculated, for example, as described below based on images (A image and B image) having an acquired parallax as in each embodiment. The calculation of the defocus amount includes, for example, the processing of dividing the image (A image and B image) into small blocks. The small block may be set for an area of a predetermined size centered on a pixel, for example, in a case where each pixel of the A image as a target is set to a target pixel. In the following description, the small block will be described as being set as a square area of $m \times m$ pixels, but the shape and size of the small block are not limited to this example.

(52) For example, when a small block is set for each pixel of the A image and the B image, correlation calculation processing is performed for each pixel (target pixel) between the two images, and a shift amount (image shift amount) of the image included in the small block corresponding to the pixel is calculated. A correlation amount $C(k)$ is calculated by the following expression (2) in the correlation calculation processing.

$$(53) \quad C(k) = \frac{1}{m} \sum_{n=1}^m E(n) \cdot F(n+k) \quad (2)$$

where m is the number of data (the number of pixels) of the (pair of) small blocks determined for the target pixel at the same position in the A image and the B image, $E(1)$ to $E(m)$ are pixel data of the pair of small blocks, k (integer) is a (data) shift amount [pixel], a Σ operation is performed for n , and n and $n+k$ are limited to a range of 1 to m . The shift amount k is a relative shift amount in the unit of the detection pitch of the pair of image data. Thus, when a correlation amount is calculated for a pair of pupil-division images (a pair of small blocks) relating to one target pixel, the shift amount k and the correlation amount $C(k)$ show a relationship illustrated in a graph of FIG. 5A, for example. At this time, the correlation amount $C(k)$ becomes minimum in the image shift amount having the highest correlation. Thereby, a shift amount x that gives the minimum value $C(x)$ to the continuous correlation amount can be calculated from the following expressions (3) and three-point interpolation method:

$$(54) \quad x = k_j + D / \text{SLOP} \quad (3) \quad C(x) = C(k_j) - \frac{D}{2} \cdot \text{SLOP} \quad (4)$$

$$\text{SLOP} = \text{MAX}\{C(k_j + 1) - C(k_j), C(k_j - 1) - C(k_j)\}$$

where k_j is a shift amount k that minimizes the discrete correlation amount $C(k)$. The shift amount x thus obtained is obtained as the image shift amount in one target pixel. A defocus amount DEF in each target pixel can be calculated from the following expression (4) using the image shift amount x :

$$(55) \quad \text{DEF} = KX \cdot PY \cdot x \quad (4)$$

where PY is a pixel pitch of the image sensor (distance between pixels in an image sensor, unit [mm/pixel]), and KX is a conversion coefficient determined by a size of an opening angle of the center of gravity of a light beam passing through the pair of distance measurement pupils. Since the size of the opening angle of the center of gravity of the light beam passing through the pair of distance measurement pupils changes according to the F-number (F_{no}) of the lens **103**, it is determined according to setting information during imaging. The F-number is a value ($F_{no}=f/A$)

obtained by dividing the focal length f of the lens by the effective aperture diameter A of the diaphragm. Thus, the defocus amount of the object in each pixel of the captured image can be calculated by repetitively calculating while shifting the position of the target pixel by one pixel each.

(56) FIG. 5B illustrates an imaging relationship between the object side and the image sensor (image plane side) using the lens **103** as a boundary in a case where the lens **103** is approximated to a thin lens. Based on the thin-lens approximation, the relationship of the following expression (5) is established:

$$(57) \quad 1/Z_o + 1/S_o = 1/f \quad (5) \quad 1/Z_{def} + 1/(S_o + DEF) = 1/f$$

where Z_o is an object-side distance of an in-focus object, Z_{def} is an object-side distance of a defocus object, S_o is an image-plane-side distance of the in-focus object, and S_{def} is an image-plane-side distance of the defocus object ($S_{def}=S_o+DEF$). From these expressions, a distance to the object can be obtained from the following expression (6):

$$(58) \quad Z_{def} = (S_{def} \cdot \text{Math. } f) / (S_{def} - f) \quad (6)$$

(59) Regarding an image shift amount, the relationship of the following expression (7) is established:

$$(60) \quad PY \cdot \text{Math. } x = \text{Math. } S_o - S_{def} \cdot \text{Math. } \text{Math. } f^2 / \{S_o \cdot \text{Math. } F_{no} \cdot \text{Math. } (S_{def} - f)\} \quad (7)$$

(61) The image shift amount is a discrete amount based on the pixel pitch PY . That is, the distance measurement resolution is determined by the sensitivity of the image shift amount. As illustrated in the following expression (7), the longer the focal length f is and the smaller the F_{no} is, the larger the minimum resolution is. Assume that the vicinity of the distance measurement target is focused, the minimum distance measurement resolution $d_{sub.TOF}$ can be approximated by the following expression (8):

$$(62) \quad d_{DEF} = DOF / 2 \quad (8) \quad DOF = 2 \cdot \text{Math. } Z_o^2 \cdot \text{Math. } F_{no} \cdot \text{Math. } PY / f^2$$

where DOF (Depth of Field) is equivalent with a depth of field corresponding to the image shift amount of 1 [pixel].

First Embodiment

(63) Referring now to FIGS. 6A to 6D and 7A to 7D, a description will be given of a method of providing sensing information by the wearable terminal **100** according to a first embodiment of the present invention. FIGS. 6A to 6D explain a system that displays sensing information by the wearable terminal **100** in this embodiment according to a work target. In FIG. 6A, a worker (user) **1001** wears the wearable terminal **100**. The wearable terminal **100** can send and receive information to and from the database server **1010** that stores three-dimensional data of work target objects **1002** and **1003** via the communication unit **54**. A data space **1011** represents a data space of the database server **1010**, and three-dimensional data (reference information) **1020** and **1030** are, for example, three-dimensional CAD data of the work target objects **1002** and **1003**, respectively. The worker **1001** is gazing at an area (gaze area, target area) **1006** indicated by dotted lines on the work target object **1003**. An object coordinate system **1007** is a coordinate system fixed to the work target object space. A camera coordinate system **1008** is a coordinate system fixed to the wearable terminal **100**.

(64) FIGS. 7A and 7B illustrate flowcharts of a method of providing sensing information by the wearable terminal **100** according to this embodiment. Each step in the flowchart in FIGS. 7A and 7B is implemented by the system control unit **50** in the wearable terminal **100** that has loaded the program stored in the nonvolatile memory **56** in the system memory **52** and executed the program to control each functional block. FIG. 7A is a flowchart illustrating a method of associating (tagging) the sensing information by the wearable terminal **100** with each target object and of storing it in the database server **1010**. FIG. 7B is a flowchart illustrating a method of comparing the sensing information by the wearable terminal **100** with the reference value stored in the database

server **1010** and of switching presented information. A description will now be given of each flowchart.

(65) First, in the step **S101**, the system control unit **50** stores in the memory **32** the captured image obtained by the imaging unit **22**. Next, in the step **S102**, the system control unit **50** extracts feature points from the captured image data using the feature-point extracting unit **33**. Next, in the step **S103**, the system control unit **50** reads out the feature point information on the past frames stored in the memory **32**, and transmits it with the newly extracted feature points to the feature-point tracking unit **34**. The feature-point tracking unit **34** compares feature points between frames. Next, in the step **S104**, the system control unit **50** transmits the matching result to the three-dimensional shape restoring unit **35**. The three-dimensional shape restoring unit **35** converts the matched feature points into three-dimensional points data.

(66) Next, in the step **S105**, the system control unit **50** receives the three-dimensional data **1030** of the work target object from the database server **1010**. The system control unit **50** aligns the three-dimensional points data and the three-dimensional data **1030** with each other using the three-dimensional shape matching unit **36**, and matches the position and the orientation. Next, in the step **S106**, the system control unit **50** estimates the position and orientation of the wearable terminal **100** in the object coordinate system **1007** (equivalent with the position and orientation in the camera coordinate system **1008**) based on the matching result in the step **S105**. At this time, grasping the position and orientation of the wearable terminal **100** in the object coordinate system **1007** can identify each of a plurality of objects having the same shape.

(67) Next, in the step **S107**, the system control unit **50** measures the dimension of the work target object **1003** using the dimension measuring unit **39**. Next, in the step **S108**, the system control unit **50** detects a scratch/dirt on the surface of the work target object **1003** using the scratch/dirt detecting unit (scratch detecting unit, dirt detecting unit) **37** and inside the work target object **1003** which is detectable by a transmission of the infrared light. Next, in the step **S109**, the system control unit **50** measures a temperature distribution on the work target object **1003** using the temperature measuring unit **38**. Next, in the step **S110**, the system control unit **50** measures the sensing time (time of acquiring the detection information) in the steps **S107**, **S108**, and **S109** using the system timer **53**.

(68) Next, in the step **S111**, the system control unit **50** transmits and stores the sensing result (detection information) to the database server **1010**. The database server **1010** stores the sensing result, such as a table illustrated in FIG. **8**. The database server **1010** stores the identification object ID (identification information), the coordinate of the sensed position (detected position) in the object coordinate system **1007**, and the sensing result and the sensing time of the steps **S107** to **S109** by associating (tagging) them with each other. For example, the dimension information may include a difference value (difference information) from the dimensional reference value of the three-dimensional data **1030**. For example, the information on the scratch or dirt may be stored with the result of presence or absence, and when the presence is determined, its shape may also be stored. The shape is measured using the dimension measuring unit **39**.

(69) Next, in the step **S112**, the system control unit **50** determines whether or not to stop the operation based on the detection result of the power switch **71** or the detection result of the operating unit **70**. If it is determined to be YES, the sensing processing flow is stopped. If it is determined to be NO, the flow returns to the step **S101** and the sensing processing flow is repeated. In the step **S113**, the system control unit **50** implements the flow illustrated in FIG. **7B**. This will be described below with reference to FIG. **7B**.

(70) First, in the step **S201**, the system control unit **50** refers to the determination result of the gaze determining unit **43** and detects the gaze point of the worker **1001**. In the gaze determination, if a time period when the visual lines of the worker **1001** are fixed onto a specific portion exceeds a predetermined threshold in the object coordinate system **1007**, it is determined that the portion is gazed. Next, in the step **S202**, the system control unit **50** identifies the object at the gaze point. In

the example illustrated in FIGS. 6A to 6D, the work target object **1003** corresponds to the target object of the gaze point.

(71) Next, in the step **S203**, the system control unit **50** requests the database server **1010** for information on the work target object **1003**. Next, in the step **S204**, the system control unit **50** receives the three-dimensional data **1030** and the table information illustrated in FIG. 8 from the database server **1010**. Next, in the step **S205**, the system control unit **50** compares the latest sensing result with the reference value. The reference value means the past information on the three-dimensional data **1030** and the table illustrated in FIG. 8.

(72) Next, in the step **S206**, the system control unit **50** determines whether or not there is a difference (feature) in the dimensional information by comparing it with the reference value. For example, in a case where the difference between the dimension value of the three-dimensional data **1030** which is the reference value and the latest dimension measurement value is equal to or more than the predetermined standard value, it is determined that there is a difference (feature). In the table illustrated in FIG. 8, it may be determined that there is a difference (feature) in a case where a difference between the latest dimensional measurement value and the past information in the last measurement that is used as the reference value is equal to or more than a predetermined standard value. In the table illustrated in FIG. 8, it may be determined that there is a difference in a case where a difference between the latest dimensional measurement value and the past information in the first measurement that is used as the reference value is equal to or more than a predetermined standard value. If it is determined to be YES, the flow proceeds to the step **S207**. If it is determined to be NO, the flow proceeds to the step **S208**.

(73) In the step **S207**, the system control unit **50** displays the dimension measuring result of the work target object **1003** on the display unit **29**. FIG. 6B illustrates a display example. A display **1110** illustrates an image displayed on the display unit **29**. A dimension display **1021** is an example in which the height dimension measurement result of the work target object **1003** does not satisfy the standard and the dimension value is displayed as information. A dimension caution display **1022** is an example of a caution display indicating how high the height dimension measurement result is in comparison with the reference value.

(74) In the step **S208**, the system control unit **50** determines whether there is a difference (feature) in the temperature information by comparing it with the reference value. For example, in a case where the latest temperature measurement result is equal to or higher than the predetermined standard value as a temperature upper limit determined for the three-dimensional data **1030**, it is determined that there is a difference (feature). In the table illustrated in FIG. 8, it may be determined that there is a difference in a case where a difference between the latest temperature measurement value and the past information in the last measurement that is used as the reference value is equal to or more than a predetermined standard value. In the table illustrated in FIG. 8, it may be determined that there is a difference in a case where a difference between the latest temperature measurement value and the past information in the first measurement that is used as the reference value is equal to or more than a predetermined standard value. It may be determined that there is a difference when a difference between the latest temperature measurement value and the past temperature information on the coordinate within a certain range measured within a certain time is equal to or more than a predetermined standard value. If it is determined to be YES, the flow proceeds to the step **S209**. If it is determined to be NO, the flow proceeds to the step **S210**.

(75) In the step **S209**, the system control unit **50** displays the temperature measurement result of the work target object **1003** on the display unit **29**. FIG. 6C illustrates a display example. A display **1110** illustrates an image displayed on the display unit **29**. A temperature display **1023** is an example in which the temperature of the work target object **1003** does not satisfy the standard and information is displayed. As illustrated in FIG. 6C, the temperature display **1023** may use a method of separating by color and displaying the temperature. A color bar display **1024** indicates the temperature (° C.) of each color.

(76) In the step **S210**, the system control unit **50** determines whether there is a difference (feature) in the scratch/dirt information by comparing it with the reference value. For example, if a shape of the detected scratch/dirt does not satisfy a permissible dimension value of the scratch/dirt determined as the reference value of the three-dimensional data **1030**, it is determined that there is a difference (feature). In the table illustrated in FIG. **8**, it may be determined that there is a difference if a difference between the shape of scratch/dirt in the latest measurement result and the past information in the last measurement that is used as a reference value is equal to or more than a predetermined standard value. In the table illustrated in FIG. **8**, it may be determined that the disappearance of the scratch/dirt is a difference if no scratch/dirt is detected in the latest measurement result using the past information in the last measurement that is used as a reference value. If it is determined to be YES, the flow proceeds to the step **S211**. If it is determined to be NO, the flow proceeds to the step **S212**.

(77) In the step **S211**, the system control unit **50** displays the scratch/dirt detection result of the work target object **1003** on the display unit **29**. FIG. **6D** illustrates a display example. A display **1110** illustrates an image displayed on the display unit **29**. A scratch display **1025** illustrates example information in which the work target object **1003** has scratches beyond the standard. The scratch display **1025** displays an area containing the scratch **1027**. A scratch caution display **1026** is a caution display example illustrating the scratch shape. In the step **S212**, the system control unit **50** determines that there is no measurement result that does not satisfy the standard, and displays no abnormal condition on the display unit **29**.

(78) The method of providing sensing information by the wearable terminal **100** has been described. This embodiment can identify a work target object required by a worker and present to the worker information on a sensing result that does not satisfy the reference value provided to the target object. Not displaying information on objects that are not work targets or unnecessary sensing information can reduce the annoying display.

(79) This embodiment compares a dimension value, temperature, scratches or dirt of the work target object with a reference value, determines if there is a difference, and displays information illustrated in the steps **S207**, **S209**, and **211** if there is the difference. For example, when the difference occurs in a plurality of sensing results, each information may be combined and displayed at the same time. For example, a priority may be previously set to a plurality of sensing results, and information only on the sensing result having a high priority may be displayed. For example, in each sensing result, the level as the degree of the difference may be configured determined and a sensing result having a high level (large difference) may be preferentially displayed.

(80) The configuration according to this embodiment receives the three-dimensional data **1030** from the database server (storage device) **1010** and stores the sensing result in the database server **1010**, but the database server **1010** is not essential. For example, all three-dimensional data and sensing result necessary for the storage device of the wearable terminal **100** may be stored. On the other hand, it is advantageous for the database server **1010** to reduce the storage capacity of the wearable terminal **100**. In addition, in a case where a plurality of wearable terminals **100** are used, there is an advantage that sensing information can be shared (detection information detected by a plurality of wearable terminals can be stored in a common storage device).

(81) The processing flows in FIGS. **7A** and **7B** in this embodiment may be carried out using a plurality of wearable terminals **100**. A sensing result of another wearable terminal **100** may be used as a reference value.

(82) This embodiment matches, in the step **S105**, the three-dimensional data **1030** with the three-dimensional points data, but the three-dimensional data **1030** is not always necessary. The three-dimensional points data acquired in the past may be used as the reference value. The three-dimensional restoring method is not limited to the method of generating the three-dimensional points data. The three-dimensional restoring method may be used while a feature amount is not set to a point but set to a geometric amount such as a line or a circle. A three-dimensional restoring

method using all pixels in the captured image data may be used, such as Dens SLAM.

(83) This embodiment uses the method of displaying the dimension, temperature, or scratch/dirt as sensing information, but the method is not limited to this example. Any sensing information may be used as long as the presentation of a plurality of sensing information is switched based on difference information from the reference value (reference information).

(84) This embodiment uses the method of storing information based on the table illustrated in FIG. 8, but the method is not limited to this example. As long as the past sensing information and the latest sensing information are compared with each other, the method is not limited.

(85) In this embodiment, the reference value is received from the database server **1010** via the communication unit **54**. However, the wearable terminal **100** itself may store the three-dimensional data and past sensing information in the memory **32** or like, and the wearable terminal **100** may read the reference value from the memory to refer to the reference value.

(86) This embodiment identifies a gazed object using the gaze detecting unit, but the method is not limited to this example. The work target object may be specified by a selection of a certain operating unit.

Second Embodiment

(87) Referring now to FIGS. 9A to 9C and 10A and 10B, a description will be given of a method of providing sensing information by a wearable terminal **100** according to a second embodiment of the present invention. The first embodiment has discussed the configuration using a single wearable terminal **100**, but the present embodiment will discuss a method using a plurality of wearable terminals **100** of providing a worker with information on a work area at a blind spot that cannot be seen from one of the wearable terminals **100**. In the first embodiment, the presented information on the work target is information on the target that the user is viewing. Therefore, for example, in the inspection work, any objects to be inspected at a blind position for the worker may be overlooked in the first embodiment. This embodiment will describe a configuration capable of providing information on a blind area of a target work target for one user which is obtained by another user, in the case where a plurality of users use the wearable terminals **100**.

(88) FIGS. 9A to 9C are explanatory diagrams of a system that supports a user having a blind spot using two wearable terminals **100a** and **100b** according to this embodiment. In FIGS. 9A to 9C, a worker (user) **1001a** as a main worker is mounted with a wearable terminal **100a**. A worker (user) **1001b** as a sub-worker is mounted on a wearable terminal **100b**. The wearable terminals **100a** and **100b** can transmit and receive information including image data to and from each other via the respective communication units **54**. The worker **1001a** is gazing at an area **1004** indicated by a dotted line on a work target object **1002**. A blind area **1005** drawn by a dotted line on the work target object **1002** is an area on the backside of the work target object **1002** when viewed from the worker **1001a**, and is a blind spot for the worker **1001a**. A display **1110a** represents a display image displayed on the display unit **29** of the wearable terminal **100a**. A display **1110b** is a display image (captured image) displayed on the display unit **29** of the wearable terminal **100b**. The object coordinate system **1007** is a coordinate system fixed on the work target object space. A camera coordinate system **1008a** is a coordinate system fixed to the wearable terminal **100a**. A camera coordinate system **1008b** is a coordinate system fixed to the wearable terminal **100b**.

(89) FIGS. 10A and 10B illustrate flowcharts of a method of supporting a user having a blind spot using two wearable terminals **100a** and **100b** according to this embodiment. Each step in the flowcharts in FIGS. 10A and 10B is implemented, when the system control unit **50** in each wearable terminal **100** loads and executes the program stored in the nonvolatile memory **56** into the system memory **52**, and controls each functional block. The system control unit **50** in each of the wearable terminals **100a** and **100b** repetitively executes the steps **S101** to **S112** in the first embodiment (FIGS. 7A and 7B). In a case where the system control unit **50** in the wearable terminal **100a** determines, in the step **S113**, that a blind-spot support display is requested via the operating unit **70**, the processing flows in FIGS. 10A and 10B are started.

(90) A description will now be given of the steps S301 to S311 of the wearable terminal 100a. Since the step S301 is the same as the step S201, a description thereof will be omitted. Next, in the step S302, the system control unit 50 identifies which object is located at the gaze point. In the example illustrated in FIGS. 9A to 9C, it is determined that the area 1004 on the work target object 1002 is gazed at, and the work target object 1002 is determined to be the gaze target object. Next, in the step S303, the system control unit 50 requests the database server 1010 for information on the work target object 1002.

(91) Next, in the step S304, the system control unit 50 receives three-dimensional data 1020 from the database server 1010. The system control unit 50 aligns the three-dimensional points data and the three-dimensional data 1020 with each other using the three-dimensional shape matching unit 36, and matches the position and the orientation. Next, in the step S305, the system control unit 50 estimates the position and orientation of the wearable terminal 100a in the object coordinate system 1007 based on the matching result of step S304. The blind area 1005 of the wearable terminal 100a is calculated based on the geometrical relationship between the position and orientation of the work target object 1002 and the position and orientation of the wearable terminal 100a. Next, in the step S306, the system control unit 50 transmits the coordinate information on the blind area 1005 in the object coordinate system 1007 and the captured image information on the wearable terminal 100a to the wearable terminal 100b.

(92) Next, in the step S307, the system control unit 50 receives the result of whether or not the blind area 1005 has been detected from the wearable terminal 100b, and the image information if it has been detected. At this time, the system control unit 50 further receives the sensing result of the blind area 1005 by the wearable terminal 100b (corresponding to the result acquired in the steps S107, S108, and S109). Next, in the step S308, the system control unit 50 determines whether there is a proper detection result of the blind area 1005 from the wearable terminal 100b. If it is determined to be YES (detection OK), the flow proceeds to the step S309. If it is determined to be NO (detection NG), the flow proceeds to the step S311.

(93) In the step S309, the system control unit 50 displays on the display unit 29 an image including the blind area 1005 received from the wearable terminal 100b. FIG. 9A illustrates a display example. In FIG. 9A, a display 1110a illustrates an image displayed on the display unit 29 of the wearable terminal 100a. Reference numeral 1002a denotes a work target object 1002 imaged by the wearable terminal 100a, and reference numeral 1003a denotes a work target object 1003. Reference numeral 1004a denotes the (gaze) area 1004 viewed from the wearable terminal 100a. Reference numeral 1120a denotes an image captured by and received from the wearable terminal 100b. In the received image 1120a, reference numeral 1005 denotes a blind area 1005 highlighted by bevel lines.

(94) Next, in the step S310, the system control unit 50 carries out the steps S205 to S212 in the first embodiment (FIGS. 7A and 7B) for the blind area 1005. At this time, the sensing result that is used in the step S205 is the sensing result by the wearable terminal 100b received in the step S307.

(95) In the step S311 the system control unit 50 gives caution on the display unit 29 that the wearable terminal 100b cannot detect the blind area 1005.

(96) Next, the steps S401 to S406 of the wearable terminal 100b will be described. The flow of the steps S401 to S406 is started in a case where the system control unit 50 of the wearable terminal 100b determines, in the step S113, that the blind-spot supporting imaging is requested via the communication unit 54.

(97) First, in the step S401, the system control unit 50 receives the coordinate information on the blind area 1005 in the object coordinate system 1007 and the captured image information on the wearable terminal 100a. Next, in the step S402, the system control unit 50 converts the received coordinate information on the blind area 1005 into the coordinate on the camera coordinate system 1008b. The coordinate transformation is similar to the step S305, and performed through a process of calculating the geometrical relationship between the position and orientation of the work target

object **1002** and the position and orientation of the wearable terminal **100b**. Next, in the step **S403**, the system control unit **50** determines whether or not the blind area **1005** is included in an imaging field of view of the wearable terminal **100b** based on the coordinate information on the blind area **1005** on the camera coordinate system **1008b**. If it is determined to be YES (the blind area **1005** is included in the imaging field of view of the wearable terminal **100b**), the flow proceeds to the step **S404**. On the other hand, if it is determined to be NO (the blind area **1005** is located outside the imaging field of view of the wearable terminal **100b**), the flow proceeds to the step **S406**.

(98) In the step **S404**, the system control unit **50** displays and highlights the blind area **1005** on the display unit **29**. FIG. **9B** illustrates a display example. In FIG. **9B**, a display **1110b** is an image (captured image) displayed on the display unit **29** of the wearable terminal **100b**. Reference numeral **1002b** denotes a work target object **1002** captured by the wearable terminal **100b**, and reference numeral **1003b** denotes a work target object **1003** captured by the wearable terminal **100b**. Reference numeral **1005b** denotes a blind area **1005** viewed from the wearable terminal **100b**. In the display example, the blind area **1005** is highlighted by bevel lines. Reference numeral **1120b** denotes an image captured by and received from the wearable terminal **100a**. In the received image **1120b**, reference numeral **1004a** denotes the (gaze) area **1004**.

(99) Next, in the step **S405**, the system control unit **50** transmits the result of the detection presence of the blind area **1005** and the captured image including the blind area **1005** to the wearable terminal **100a**. At this time, the sensing information on the blind area **1005** by the wearable terminal **100b** is transmitted to the wearable terminal **100a** in the form of the table illustrated in FIG. **8**.

(100) In the step **S406**, the system control unit **50** transmits the result of the detection absence of the blind area **1005** to the wearable terminal **100a**.

(101) The method of providing sensing information by the wearable terminals **100a** and **100b** according to this embodiment has been described. Using the plurality of wearable terminals **100a** and **100b**, this embodiment can support a certain operator having a blind area, and can present information on the sensing result of the blind area. Receiving the image information on the blind area from the other wearable terminal **100** and displaying it on the display unit **29** can prevent the blind spot from being overlooked in the work.

(102) In this embodiment, in the step **S306**, the wearable terminal **100a** converts the coordinate of the blind area **1005** into the object coordinate system **1007** and then transmits it to the wearable terminal **100b**, but the method is not limited to this method. For example, the wearable terminal **100b** may receive the coordinate of the blind area **1005** in the camera coordinate system **1008a** and perform the coordinate conversion. Alternatively, the wearable terminal **100b** may receive the captured image information including the blind area **1005** from the wearable terminal **100a** and perform the coordinate conversion based on the captured image information. Moreover, the database server **1010** may perform the coordinate conversion between the object coordinate system **1007** and each camera coordinate system.

(103) This embodiment transmits, in the step **S406**, the fact that the blind area **1005** is undetectable to the wearable terminal **100a**, but information on a positional relationship between each wearable terminal **100** and the target object may be created and shared. FIG. **11** is a display example of positional relationship information shared by the wearable terminals **100a** and **100b**. For example, a map may be displayed so that the wearable terminals **100a** and **100b** and the work target objects **1002** and **1003** can be viewed from above. In FIG. **11**, a dotted line illustrated by **1220** represents an imaging viewing angle of the wearable terminal **100b**, and clearly indicates that the blind area **1005** is out of the field of view. Thereby, the worker **1001b** can recognize a direction in which the blind area **1005** can be imaged. The worker **1001a** can recognize whether or not there is another worker who can image the blind area **1005** nearby.

(104) This embodiment does not need the database server **1010**. For example, in one method, the three-dimensional data and the sensing result required for the storage device of each wearable

terminal **100** are distributed and stored, and the necessary data may be received from the wearable terminal **100** possessing it.

(105) This embodiment identifies the gaze object using the gaze detecting unit, but is not limited to this method. The work target object may be specified by a selection of a certain operating unit.

(106) This embodiment uses the same hardware configuration for both the wearable terminals **100a** and **100b**, but may use different hardware as long as the hardware configuration can support the blind spot. For example, a handheld camera instead of a wearable terminal may be used.

Alternatively, an autonomous mobile terminal, such as a drone and a mobile robot, may be used.

(107) This embodiment uses two terminals, the wearable terminals **100a** and **100b**, but the present invention is not limited to this configuration, and may use any number of terminals. Applying this embodiment can transmit the captured image of the terminal that could capture the blind area to the terminal of the main worker. This configuration can provide the effect of reducing a communication amount between a plurality of terminals. It is annoying to display imaging information on all terminals on the terminal of the main worker. Applying this embodiment can transmit only the captured image of the terminal that could capture the necessary blind area, and the annoyance can be reduced.

Third Embodiment

(108) Referring now to FIGS. **12**, **13A**, and **13B**, a description will be given of a method of providing sensing information by the wearable terminal **100** according to a third embodiment of the present invention. As illustrated in the external view illustrated in FIG. **1**, the TOF detecting unit (second distance measurement unit) **83** is disposed outside both lens optical systems of the imaging units **150R** and **150L**, and has an angle of view wider than that of the imaging unit (first distance measurement unit) **22**. Its detection range covers several meters, and the distance measurement resolution depends on the distance to the measurement target. In the imaging-plane phase-difference distance measurement method of the imaging unit **22**, the distance measurement accuracy varies depending on the lens parameter represented by the optical zoom magnification in each of the imaging units **150R** and **150L**. This embodiment will discuss a method for improving the distance measurement accuracy in a configuration mounted with a distance measurement sensor depending on a distance measurement distance and a distance measurement sensor depending on a lens mechanism.

(109) FIG. **12** is a flowchart of a dimension measuring method by the wearable terminal **100** according to this embodiment. Each step in the flowchart illustrated in FIG. **12** is implemented by the system control unit **50** in the wearable terminal **100** loading the program stored in the nonvolatile memory **56** into the system memory **52**, executing the program, and controlling each functional block. The system control unit **50** in the wearable terminal **100** repetitively executes the flow of the steps **S101** to **S112** in the first embodiment (FIGS. **7A** and **7B**). In the step **S107**, when the system control unit **50** measures the dimension of the work target object **1003** using the dimension measuring unit **39**, the processing flow illustrated in FIG. **12** is started.

(110) Since the step **S501** is the same as the step **S201**, a description thereof will be omitted. Next, in the step **S502**, the system control unit **50** executes AF processing for focusing on the gaze point by moving the focus lens in the lens **103** via the AF driving circuit **3**. Next, in the step **S503**, the system control unit **50** performs AE processing on the gaze point and controls the aperture in the diaphragm **1** so that the exposure in the imaging unit **22** has a proper amount. At this time, the exposure time of the imaging unit **22** and the conversion gain in the photoelectric conversion may be changed. Next, in the step **S504**, the system control unit **50** irradiates infrared light using the infrared LED illumination unit (second distance measurement unit) **81**, measures the time lag of the reflected light using the TOF detecting unit **83**, and calculates a distance D_{TOF} to the object near the gaze point.

(111) Next, in the step **S505**, the system control unit **50** calculates the distance measurement reliability (second reliability) α_{TOF} by the TOF detecting unit **83** as illustrated in the following

expression (9):

$$(112) \quad \alpha_{\text{TOF}} = k1 \cdot \text{Math.} (d_{\text{Tmax}} - d_{\text{TOF}}) / d_{\text{Tmax}} \quad (9)$$

where **k1** is a constant, **d.sub.TOF** is a distance measurement resolution illustrated in the expression (1), **d.sub.Tmax** is a maximum distance measurement resolution, and **d.sub.TOF** has a value equal to or lower than **d.sub.Tmax** within the effective distance measurement range. The number of effective signal electrons **N** illustrated in the expression (1) depends on the distance of the object. For example, a relationship between the object distance and the number of effective signal electrons **N** may be tabulated and stored in the nonvolatile memory **56** in advance.

(113) Next, in the step **S506**, the system control unit **50** calculates the distance **D_DEF** to the object at or near the gaze point by the imaging-plane phase-difference distance measurement method of the imaging unit **22**. Next, in the step **S507**, the system control unit **50** calculates the distance measurement reliability (first reliability) α_{DEF} by the imaging-plane phase-difference distance measurement method of the imaging unit **22** as illustrated in the following expression (10):

$$(114) \quad \alpha_{\text{DEF}} = k2 \cdot \text{Math.} (d_{\text{Dmax}} \cdot \text{Math.} d_{\text{DEF}}) / d_{\text{Dmax}} \quad (10)$$

where **k2** is a constant, **d.sub.DEF** is a distance measurement resolution illustrated in the expression (8), **d.sub.Dmax** is a maximum distance measurement resolution, and **d.sub.DEF** has a value equal to or lower than **d.sub.Dmax** within a range of the effective defocus amount.

(115) Next, in the step **S508**, the system control unit **50** calculates a distance measurement result **Depth** for the target object by the following expression (11):

$$(116) \quad \text{Depth} = D_{\text{DEF}} \cdot \text{Math.} \alpha_{\text{DOC}} / (\alpha_{\text{DEF}} + \alpha_{\text{TOF}}) + D_{\text{TOF}} \cdot \text{Math.} \alpha_{\text{TOF}} / (\alpha_{\text{DEF}} + \alpha_{\text{TOF}}) \quad (11)$$

(117) The expression (11) is equivalent with setting each distance measurement reliability illustrated in the expressions (9) and (10) to a weight and calculating an internal division point of the distance measurement result by both distance measurement units. The distance measurement result reflecting each reliability can be obtained by the expression (11).

(118) Next, in the step **S509**, the system control unit **50** transmits the **Depth** measurement result to the dimension measuring unit **39**, and measures the dimension of the measurement target. FIGS. **13A** and **13B** explain the distance measurement reliabilities of the two distance measurement units mounted on the wearable terminal **100**. Reference numeral **1301** in FIG. **13A** is a graph illustrating a relationship between the distance **Z** from the TOF detecting unit **83** to the distance measurement target and the distance measurement reliability α_{TOF} . Assume that the distance measurement reliability α_{TOF} is valid at a distance between the minimum measurement distance **Z.sub.min**, and the maximum measurement distance **Z.sub.max**, and is invalid at other distances ($\alpha_{\text{TOF}}=0$). The distance measurement reliability α_{TOF} has a higher value as the distance measurement target is closer to the minimum measurement distance **Z.sub.min**, and has a lower value as the distance measurement target is farther.

(119) Reference numeral **1302** in FIG. **13B** is a graph illustrating a relationship between the depth of field **DOF** of the lens **103** and the distance measurement reliability α_{DEF} . Assume that the distance measurement reliability α_{DEF} is valid at a depth of field between the minimum depth of field **DOF.sub.min** and the maximum depth of field **DOF.sub.max**, and is invalid at other depths of field ($\alpha_{\text{DEF}}=0$). The distance measurement reliability α_{DEF} has a higher value when the depth of field at the position of the distance measurement target is shallower than the minimum depth of field **DOF.sub.min**, and has a lower value when the depth of field is deeper. For example, in a case where the distance measurement target is relatively close, the distance measurement reliability α_{TOF} has a high value.

(120) When the optical zoom magnification of the lens **103** is lowered in order for the user to view a close position, the depth of field **DOF** becomes deeper and the distance measurement reliability α_{DEF} has a lower value. As a result, a reference ratio of **D_TOF** is increased in the **Depth** calculation. For example, in a case where the distance measurement target is located at a relatively

distant position, the distance measurement reliability α_{TOF} has a low value. On the other hand, when the optical zoom magnification of the lens **103** is increased in order for the user to view a distant position, the depth of field DOF becomes shallower and the distance measurement reliability α_{DEF} has a higher value. As a result, a reference ratio of D_{DEF} is increased in the Depth calculation. Thus, the expression (11) can improve the distance measurement accuracy by taking advantage of both distance measurement methods according to the position of the distance measurement target.

(121) In this embodiment, the wearable terminal **100** includes a first distance measurement unit (imaging unit **22**) and a second distance measurement unit (infrared LED illumination unit **81**, TOF detecting unit **83**). The wearable terminal **100** further includes a first calculating unit, a second calculating unit, and a control unit (system control unit **50**) that serves as a third calculating unit. The first calculating unit calculates the first reliability (first distance measurement reliability) regarding the first distance measurement result by the first distance measurement unit. The second calculating unit calculates the second reliability (second distance measurement reliability) regarding the second distance measurement result by the second distance measurement unit. The third calculating unit calculates a distance measurement result by changing a reference ratio between the first distance measurement result and the second distance measurement result, based on the first reliability and the second reliability. The first reliability is lower as the depth of field of the lens unit at the distance measurement target position is deeper. The second reliability is lower as the position of the distance measurement target is farther from the second distance measurement unit.

(122) In this embodiment, the wearable terminal **100** may include a determining unit (gaze determining unit **43**) for determining the gaze area (area **1006**) that the user is gazing at. The first distance measurement unit enables the lens unit (lens **103**) to be focused on the gaze area, and performs a distance measurement at the in-focus target position. The second distance measurement unit may contain the first distance measurement angle of view of the first distance measurement unit and is wider than the first distance measurement angle of view.

(123) The method of providing the sensing information by the wearable terminal **100** according to this embodiment has been described. This embodiment can improve the distance measurement accuracy by integrating the output results from a plurality of distance measurement units by the wearable terminal **100**.

(124) The SNR of the TOF detecting method depends on the reflected light amount of the distance measurement target. The number of effective signal electrons N represented by the expression (1) is affected not only by the distance but also by the reflectance of the object and the ambient light. Therefore, the distance measurement reliability α_{TOF} may have a value that depends on the object and the scene. For example, the object may be identified by deep learning from the captured image by the imaging unit **22**, and an estimated value of the number of effective signal electrons N may be changed according to the reflectance of the object. Similarly, the scene may be identified by deep learning, and the estimated value of the number of effective signal electrons N may be changed, for example, by a difference between a day scene and a night scene.

(125) The TOF detecting method is bad at detecting transparent or translucent objects. For example, if it is determined to be a transparent object based on the object identification result, a value of α_{TOF} may be set to zero. That is, in the case where the object is a transparent object or a translucent object, the control unit (system control unit **50**) may set the second reliability to zero. From the expression (11), only the result of the imaging-plane phase-difference distance measurement method is referred to, and an erroneous detection result for the transparent object by the TOF can be avoided.

(126) In this embodiment, the control unit may set the second reliability to zero in a case where the distance to the distance measurement target is shorter than the shortest measurable distance of the second distance measurement unit, or in a case where the distance to the distance measurement

target is longer than the longest measurable distance of the second distance measurement unit. The control unit may set the first reliability to zero in a case where the depth of field of the lens unit is shallower than the minimum depth of field of the first distance measurement unit, or in a case where the depth of field of the lens unit is deeper than the maximum depth of field of the first distance measurement unit.

(127) The distance measurement accuracy by the imaging-plane phase-difference distance measurement method depends on the focal length of the lens **103** and the aperture in the diaphragm as illustrated in the expression (8). For example, a lower limit threshold may be set for the value of α_{DEF} , and the focal length and the aperture in the diaphragm may be changed in a case where the value is lower than the lower limit threshold. More specifically, in the case where the value of α_{DEF} is lower than the lower limit threshold, the value of α_{DEF} can be improved by increasing the optical zoom magnification. If an exposure amount does not reach a saturation upper limit value of the imaging unit **22**, the value of α_{DEF} can be improved by an opening control of the aperture in the diaphragm. That is, in this embodiment, the wearable terminal **100** may include a zoom changing unit (lens system control circuit **4**) for changing the zoom magnification of a zooming unit in the lens **103**, and a diaphragm changing unit (diaphragm driving circuit **2**) for changing the aperture in the diaphragm **1**. In the case where the first reliability is smaller than a predetermined threshold, the control unit controls at least one of the zoom changing unit and the diaphragm changing unit so that the first reliability becomes higher.

(128) In this embodiment, the defocus amount is generated based on, but is not limited to, images having a parallax by the image-plane phase-difference distance measurement method (phase-difference detecting method). For example, a Depth from Defocus (“DFD”) method may be used in which the defocus amount is calculated from the correlation between two images having different focus and F-number. Alternatively, a Depth from Focus method may be used. The present invention can be similarly implemented regardless of the defocus amount obtained by either method.

(129) This embodiment uses the distance measurement result by the TOF detecting unit **83**, but the present invention is not limited to this method. For example, a focusing detecting method may use a parallax between two images such as a stereo camera. Another method may be a method of emitting ultrasonic waves to an object like a sonar and performing a distance measurement from a time difference of the reflected sound waves. Still another method is an active stereo method that irradiates geometrically patterned light and performs a distance measurement from a geometric shape of the reflection pattern. That is, as long as the distance measurement performance depends on the distance, this embodiment is similarly applicable.

Fourth Embodiment

(130) Referring now to FIGS. **14** to **18**, a description will be given of a method of controlling the display sensitivity by the wearable terminal **100** according to a fourth embodiment of the present invention. The control of the display sensitivity relates to a control over a display angle of view when the object is enlarged, displayed, and visually recognized using the wearable terminal **100**.

(131) FIG. **14** explains a relationship between the imaging angle of view and the display angle of view of the wearable terminal **100** in this embodiment, and illustrates a relationship between the imaging unit **22** and the display unit **29** when a user **2000** visually recognizes the object at a telephoto position. An angle of view (captured image) **2003** of the imaging unit **22** indicates an angle of view of the imaging unit **22**, and images an object **2001** and a peripheral area of the object **2001**. An angle of view **2004** of the display unit **29** illustrates an angle of view of a display image obtained by cutting out part or whole of the image captured by the imaging unit **22** by the image processing unit **24** and displaying on the display unit **29** at a specific magnification. The object image **2002** in the angle of view **2004** of the display unit **29** is an enlarged display at a specific magnification of the object **2001** in the angle of view **2003** of the imaging unit **22**.

(132) FIG. **15** is a bird's-eye view of FIG. **14** viewed from the Y-axis direction. A description of the elements designated by the same reference numerals as those in FIG. **14** will be omitted. FIG. **15** is

a display in a case where a specific magnification displayed on the display unit **29** is set to α times. Although the imaging unit **150L** is taken as an example, the control over the imaging unit **150R** is similar. An object image **2007** at the center of the angle of view **2004** of the display unit **29** is an enlarged display at a specific magnification of an object **2006** at the center of the angle of view **2003** of the imaging unit. Now assume that θ is an angle between the object **2001** imaged on the imaging unit **150L** and the optical axis, and Φ is an angle between the center of the angle of view **2004** of the display unit **29** and the object image **2002**.

(133) In a case where the gaze point of the user **2000** is in the object image **2007** and the gaze point is moved to the object image **2002**, the user **2000** changes the direction of his head by Φ based on the display image of the display unit **29**. In the wearable terminal **100** attached to the head of the user **2000**, since the imaging unit **150L** and the display unit **29** are integrated, the direction of the imaging unit **150L** is also changed by Φ , and the optical axis is moved to a point **2005**. The user **2000** wants to visually recognize the object image **2002**, but the display unit **29** enlarges the display by a times and thus the object **2001** is consequently lost. FIG. **15** is an example illustrating the movement of the head in the horizontal direction, which is the X-axis, using a bird's-eye view viewed from the Y-axis direction, but this is similarly applicable when the head is moved in the vertical direction on the unillustrated Y-axis.

(134) FIGS. **16A** to **16F** explain control over a display angle of view for a translational movement of the wearable terminal **100**, and illustrate an image formed on the imaging unit **150L** in FIG. **15** and a display image on the display unit **29**. A description the elements designated by the same reference numerals as those in FIG. **14** will be omitted. FIG. **16A** illustrates the angle of view **2003** of the imaging unit, and illustrates a relationship between a cutout angle of view **2201** and the object **2001**. FIG. **16B** illustrates the angle of view **2004** of the display unit, which is the same as the cutout angle of view **2201** in FIG. **16A**, and as illustrated in FIG. **15**, part of the angle of view **2003** of the imaging unit is displayed and enlarged on the display unit **29** at a magnification of α times. A relationship between FIGS. **16C** and **16D** and a relationship between FIGS. **16E** and **16F** are similar to the relationship between FIGS. **16A** and **16B**.

(135) FIG. **18** is a flowchart of control over the display angle of view by the wearable terminal **100**. First, in the step **S601**, the system control unit **50** determines whether to reset a cutout range of the imaging unit **150L**. The cutout range resetting in this embodiment means returning a center coordinate of a cutout angle of view **2201** to the center of the angle of view **2003** of the imaging unit. For example, the user **2000** operates the operating unit (reset operating unit) **70** to perform the reset operation. Next, in the step **S602**, the system control unit **50** determines whether or not to reset the cutting range. If the cutting range is to be reset, the flow proceeds to the step **S606**. On the other hand, if the cutout range is not to be reset, the flow proceeds to the step **S603**.

(136) In the step **S603**, the system control unit **50** determines whether to set the display sensitivity control mode. The display sensitivity control mode is determined by the user **2000** selecting a mode in advance using the operating unit (mode selecting unit) **70**, or by the gaze determining unit (determining unit) **43** based on whether or not a specific object is gazed. If the gaze determination by the gaze determining unit **43** is used to determine whether to set the display sensitivity control mode and gazing is determined by the gaze determining unit **43**, the display sensitivity control mode is set and if gazing is not determined by the gaze determining unit **43**, no display sensitivity control mode is set.

(137) Next, in the step **S604**, the system control unit **50** determines whether or not the display sensitivity control mode is set (whether or not the display sensitivity control is performed). If the display sensitivity control is to be made, the flow proceeds to the step **S605**. On the other hand, if no display sensitivity control is performed, the flow proceeds to the step **S606** or **S607**.

(138) Referring now to FIGS. **16A** to **16F**, a description will be given of control over the display angle of view for the translational movement of the angle of view **2003** of the imaging unit **22** in the display sensitivity control in the step **S605**. As illustrated in FIG. **16D**, assume a case where the

center of the angle of view **2004** of the display unit at a first gaze point **2203** is moved to a second gaze point **2202**. In the angle of view **2004** of the display unit **29** in FIG. **16D**, Φ is an angle between the first gaze point **2203** and the second gaze point **2202**. x is a distance between the user wearing the wearable terminal **100** and the display unit **29**, and ad is a distance between the first gaze point **2203** and the second gaze point **2202** on the display unit **29**. A distance d is a distance between the first gaze point **2203** and the second gaze point **2202** on the angle of view **2003** of the imaging unit **22**, which will be described below. The distance ad is a distance obtained by multiplying the distance d by a magnification of a times. Hence, the distance ad is equal to $x \tan \Phi$, and the distance between the first gaze point **2203** and the second gaze point **2202** on the display unit **29** in the case where the user moves his head by the angle Φ can be calculated.

(139) On the other hand, as illustrated in FIG. **16C**, when viewed based on the angle of view **2003** of the imaging unit, θ is an angle between the first gaze point **2203** and the second gaze point **2202**, and d is a distance between the first gaze point **2203** and the second gaze point **2202** on the angle of view **2003** of the imaging unit **22** at this time. FIG. **16E** illustrates the angle of view **2003** of the imaging unit when the user **2000** moves his head by Φ in order to move the center of the angle of view from the first gaze point **2203** to the second gaze point **2202** based on the angle of view **2004** of the display unit **29** illustrated in FIG. **16D**. The first gaze point **2203** is moved to a position separated by ad from the center of the angle of view **2003** of the imaging unit. At this time, a cutout angle of view **2201** is moved by a distance $(\alpha-1)d$ so that the angle is θ between the center of the cutout angle of view **2201** and the first gaze point **2203**, which was the center of the angle of view **2003** of the imaging unit before the user **2000** moves his head. Moving the cutout angle of view **2201** from the center of the angle of view **2003** of the imaging unit in this way, as illustrated in FIG. **16F** can control the angle of view **2004** of the display unit so as to display the second gaze point **2202** at the center.

(140) A moving amount between the first gaze point **2203** and the second gaze point **2202** of the user has been described. In other words, a cutout position of the captured image is changed so that a moving amount of a projected point between moving imaging frames of the wearable terminal **100** and a moving amount of the projected point between display frames match with respect to an arbitrary projected point of the object. Here, the moving amount of the projected point between the imaging frames means a moving amount in the coordinate system normalized by the maximum number of pixels on the x-axis and the y-axis of the imaging frame. Similarly, the moving amount of the projected point between the display frames means a moving amount in the coordinate system normalized by the maximum number of pixels on the x-axis and the y-axis of the display frame.

(141) In the example of FIG. **16E**, the cutout angle of view **2201** is the cutout position near the left end of the angle of view **2003** of the imaging unit, and there is little room for moving the cutout position to the left. In this case, as illustrated in FIG. **16F**, a cutout position caution **2204** is displayed at the right end of the angle of view **2004** on the display unit, or a direction indicating bar (auxiliary display) **2205** is displayed to give a caution notification of the display sensitivity control limit to the user **2000**. Alternatively, in a case where the imaging unit **150L** has an optical zoom (not shown), the object image **2002** may be controlled so as not to deviate from the angle of view **2003** of the imaging unit by controlling the optical zoom to the wide-angle side.

(142) Referring now to FIGS. **17A** to **17D**, a description will be given of control over the display angle of view against a rotation of the angle of view **2003** of the imaging unit **22** about the optical axis in the display sensitivity control in the step **S605** illustrated in FIG. **18**. FIGS. **17A** to **17D** explain control over the display angle of view against the rotation about the optical axis of the wearable terminal **100**. FIG. **17A** illustrates the angle of view **2003** of the imaging unit, and illustrates a relationship between an angle of view center **2301** of the imaging unit, a cutout angle of view **2201**, and a first coordinate point **2302**. The first coordinate point **2302** is located at the center of the cutout angle of view **2201** and corresponds to an angle of view center (center on the display image) **2303** of the display unit. FIG. **17B** illustrates the angle of view **2004** of the display

unit **29**, which is the same as the cutout angle of view **2201** illustrated in FIG. **17A**, and as illustrated in FIG. **15**, part of the angle of view **2003** of the imaging unit is displayed and enlarged at a magnification of a times on the display unit **29**. A relationship between FIGS. **17C** and **18D** is the same as the relationship between FIGS. **17A** and **17B**.

(143) Now assume a case where the user **2000** rotates his head by an angle β in order for the user **2000** who is viewing the angle of view **2004** of the display unit illustrated in FIG. **17B** to rotate about the angle of view center **2303** of the display unit as an axis by the angle β based on the angle of view **2004** of the display unit **29**. FIG. **17C** illustrates the angle of view **2003** of the imaging unit when the user **2000** rotates his head by the angle β in order to display and rotate about the angle of view center **2303** of the display unit **29** as an axis by the angle β .

(144) At this time, the center of the cutout angle of view **2201** is moved from the first coordinate point **2302** to a second coordinate point **2304**. The second coordinate point **2304** is an endpoint different from the angle of view center **2301** of the imaging unit when a line segment made by connecting the angle of view center **2301** of the imaging unit and the first coordinate point **2302** is rotated by an angle β about the angle of view center **2301** of the imaging unit. As a result of moving the center of the cutout angle of view **2201** to the second coordinate point **2304**, the angle of view **2004** of the display unit is obtained which is rotated by the angle β about the angle of view center **2303** of the display unit illustrated in FIG. **17D**. Similarly, in a case where the imaging unit **150L** is rotated about the angle of view center **2301** of the imaging unit, the center point of the cutout angle of view **2201** is moved on a circle **2305** having a radius from the first coordinate point **2302** of the imaging unit to the angle of view center **2303**.

(145) In the step **S606** illustrated in FIG. **18**, an image having a specific magnification is displayed on the display unit **29** while the center of the cutout angle of view **2201** at the angle of view **2003** of the imaging unit **22** and the angle of view center **2301** of the imaging unit are always accorded with each other. In the step **S607**, display fixing control is performed for such control that an arbitrary projected point on the object at the angle of view **2003** of the imaging unit is fixed on the angle of view **2004** of the display unit **29**. The display fixing control is performed in a normal digital camera as an image stabilizing control, and a detailed description thereof will be omitted because it is a known technique.

(146) Thus, in this embodiment, the system control unit **50** may fix the cutout position to the center of the captured image in the case where the first mode is selected by the operating unit (mode selecting unit) **70** that accepts a mode selection from the user. In the case where the second mode is selected by the mode selecting unit, the control unit may change the cutout position so that the projected point displayed at each coordinate of the display image is maintained at each coordinate of the display image. The wearable terminal **100** may include a determining unit (gaze determining unit **43**) for determining whether or not the user is gazing at a specific object, and the control unit may fix the cutout position at the center of the captured image in the case where the user is not gazing at the specific object. In the case where the cutout position includes a predetermined range near the end of the captured image, the control unit may change a display color of part of the display image according to the predetermined range included in the cutout position. In the case where the cutout position includes a predetermined range near the end of the captured image, the control unit may display an auxiliary display (direction indicating bar **2205**) indicating a direction according to the predetermined range included in the cutout position. The wearable terminal **100** may include a zoom changing unit (lens system control circuit **4**) that can provide optical zooming by controlling the lens **103**. In the case where the cutout position includes the predetermined range near the end of the captured image, the control unit may perform zoom-out control by using the zoom changing unit.

(147) The method of providing sensing information by the wearable terminal **100** in this embodiment has been described. This embodiment performs the display sensitivity control that combines the control over the display angle of view against a translational movement and the

control over the display angle of view against a rotation about the optical axis as a center, and can present image information with less discomfort for the intended operation for moving the viewpoint of the user **2000**. In the display sensitivity control, a configuration may be used which enables the user **2000** to independently turn on and off the control over the display angle of view against the translational movement and the control over the display angle of view against the rotation about the optical axis as a center. In this embodiment, the image acquired by the imaging unit **22** is enlarged and displayed on the display unit **29**, but a configuration including the imaging unit **150L** and the display unit **160L** capable of displaying a wider angle range than the see-through display may provide the display sensitivity control.

(148) In each embodiment, the system control unit **50** may provide controls through single hardware or the entire apparatus may be controlled by a plurality of hardware components sharing the processes.

OTHER EMBODIMENTS

(149) Embodiment(s) of the present invention can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a ‘non-transitory computer-readable storage medium’) to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)TM), a flash memory device, a memory card, and the like.

(150) Each embodiment can provide an electronic apparatus and a control method of the same, each of which can preset proper detection information according to a state of a target object.

(151) While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

(152) While each embodiment has described a wearable terminal as an electronic apparatus, each embodiment is also applicable to an electronic apparatus other than the wearable terminal. Each embodiment may be applied, for example, to an image pickup apparatus such as a digital camera, a portable digital microscope, a display apparatus such as an image viewer, and an autonomous mobile terminal such as a drone or a mobile robot. Each embodiment may be applied to a personal computer, a PDA, a mobile phone terminal, a game machine, an electronic book reader, or the like. Each electronic apparatus can display proper sensing information according to the state of the work target, and improve the work efficiency. The above embodiments may be combined.

(153) This application claims the benefit of Japanese Patent Application No. 2021-028555, filed on Feb. 25, 2021, which is hereby incorporated by reference herein in its entirety.

Claims

1. An electronic apparatus comprising: at least one processor; and at least one memory coupled to the at least one processor storing instructions that, when executed by the at least one processor, cause the at least one processor to function as: a determining unit configured to determine a target object that a user is paying attention to for a predetermined period or longer; a detecting unit configured to detect detection information on the target object; a referring unit configured to refer to reference information associated with the target object; a presenting unit configured to present the detection information; a control unit configured to control the presenting unit to present the detection information when there is a difference equal to or more than a predetermined standard value between the reference information and the detection information; a first calculating unit configured to calculate a first reliability regarding a first distance measurement result by the first distance measurement unit; a second calculating unit configured to calculate a second reliability regarding a second distance measurement result by the second distance measurement unit; and a third calculating unit configured to calculate a distance measurement result by changing a reference ratio between the first distance measurement result and the second distance measurement result based on the first reliability and the second reliability, wherein the first reliability becomes lower as a depth of field of a lens unit provided in the first distance measurement unit at a position of a distance measurement target becomes deeper, and wherein the second reliability becomes lower as the position of the distance measurement target becomes farther from the second distance measurement unit.
 2. The electronic apparatus according to claim 1, wherein the first distance measurement unit is a distance measurement unit of a phase difference detecting type, a depth from defocus type, or a depth from focus type, and wherein the second distance measurement unit is a distance measurement unit of a time of flight type or a triangulation type.
 3. The electronic apparatus according to claim 1, wherein the at least one processor further functions as: a zoom changing unit configured to change a zoom magnification of a zooming unit in the lens unit; and a diaphragm changing unit configured to change an aperture in a diaphragm in the lens unit, wherein when the first reliability is lower than a predetermined threshold, the control unit controls at least one of the zoom changing unit and the diaphragm changing unit so that the first reliability becomes higher.
 4. The electronic apparatus according to claim 1, wherein the determining unit is configured to determine a gaze area of the target object that the user is gazing at, wherein the first distance measurement unit performs a distance measurement of the gaze area at a first angle of view, and wherein the second distance measurement unit performs a distance measurement at a second angle of view that contains and is wider than the first angle of view.
 5. An electronic apparatus comprising: an imaging unit; a display unit configured to cut out and enlarge at least part of a captured image; at least one processor; and at least one memory coupled to the at least one processor storing instructions that, when executed by the at least one processor, cause the at least one processor to function as: a determining unit configured to determine a target object that a user is paying attention to for a predetermined period or longer; a detecting unit configured to detect the target object; a referring unit configured to refer to reference information associated with the target object; a presenting unit configured to present the detection information; a control unit configured to control the presenting unit to present the detection information when there is a difference equal to or more than a predetermined standard value between the reference information and the detection information, wherein when the imaging unit is moved, the control unit changes a cutout position of the captured image so that a moving amount of a projected point of a point fixed in a space onto the captured image moved between continuous imaging frames captured while the imaging unit is moved accords with a moving amount of the projected point between display frames that enlarges and displays the projected point for each imaging frame.
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